

Differential Dynamic Programming (DDP): Forward Pass Decomposition for Copper Plate MPOPF

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1 Problem Overview

Differential Dynamic Programming (DDP) decomposes a multi-period optimal power flow (MPOPF) problem into T sequential subproblems using a forward pass approach. Each time step t solves a local optimization problem that couples with adjacent time steps through:

- **Backward coupling:** State trajectory from previous time step $t - 1$ (current iteration)
- **Forward coupling:** Shadow prices (dual variables) from next time step $t + 1$ (previous iteration)

The copper plate formulation ignores network constraints and focuses on energy arbitrage with battery storage.

2 Sets, Indices, and Parameters

2.1 Sets and Indices

- $\mathcal{T} = \{1, 2, \dots, T\}$: Set of time periods
- $t \in \mathcal{T}$: Time period index
- k : DDP iteration index

2.2 System Parameters

$$P_{\text{BASE}} = 1000 \text{ kW (power base)} \tag{1}$$

$$E_{\text{BASE}} = 1000 \text{ kWh (energy base)} \tag{2}$$

$$\Delta t = 1.0 \text{ hour (time step duration)} \tag{3}$$

2.3 Load and Price Profiles

- $P_L[t]$: Load demand at time t [pu on P_{BASE}]
- $C[t]$: Energy price at time t [\$/kWh]
- C_B : Battery quadratic cost coefficient [\$/kW²/h], typically $10^{-6} \times \min(C[t])$

2.4 Battery Parameters

$$E_{\text{Rated}} : \text{Battery energy capacity [pu on } E_{\text{BASE}}] \quad (4)$$

$$P_{B,R} : \text{Battery power rating [pu on } P_{\text{BASE}}] \quad (5)$$

$$\text{SOC}_{\min}, \text{SOC}_{\max} : \text{SOC limits (fraction of } E_{\text{Rated}}) \quad (6)$$

$$B_0 : \text{Initial SOC [pu on } E_{\text{BASE}}] \quad (7)$$

$$B_{T,\text{target}} : \text{Terminal SOC constraint (optional) [pu on } E_{\text{BASE}}] \quad (8)$$

SOC bounds:

$$\underline{B} = \text{SOC}_{\min} \cdot E_{\text{Rated}} \quad (9)$$

$$\overline{B} = \text{SOC}_{\max} \cdot E_{\text{Rated}} \quad (10)$$

3 Variable Color Coding

Following DDP forward pass decomposition:

- $\mathbf{B}[\mathbf{t}]$: Battery SOC at time t in **current iteration** k (blue)
- $\mathbf{B}[\mathbf{t}]$: Battery SOC at time t from **previous iteration** $k - 1$ (red, used for coupling)
- $\mu[\mathbf{t}]$: Shadow price (dual variable) for battery dynamics at time t (green)
- $P_{\text{subs}}[t]$: Substation power at time t [pu]
- $P_B[t]$: Battery power at time t [pu] (positive = discharge, negative = charge)

4 DDP Forward Pass Subproblem

4.1 Subproblem for Time Step t (Iteration k)

Each time step $t \in \{1, \dots, T\}$ solves the following optimization problem:

$$\begin{aligned} \min_{\substack{P_{\text{subs}}[t], P_B[t], \mathbf{B}[\mathbf{t}], \\ P_B[t+1] \text{ (if } t < T)}} \quad & \underbrace{C[t] \cdot P_{\text{subs}}[t] \cdot \Delta t}_{\text{Energy cost at } t} + \underbrace{C_B \cdot (P_B[t])^2 \cdot \Delta t}_{\text{Battery cost at } t} \\ & + \underbrace{\mu^{\mathbf{k}-1}[\mathbf{t} + \mathbf{1}] \cdot \left(\mathbf{B}^{\mathbf{k}-1}[\mathbf{t} + \mathbf{1}] - \mathbf{B}[\mathbf{t}] + \Delta t \cdot P_B[t + 1] \right)}_{\text{Soft coupling with next time step (if } t < T)} \end{aligned} \quad (11)$$

Subject to:

Power Balance:

$$P_{\text{subs}}[t] + P_B[t] = P_L[t] \quad (12)$$

Battery Dynamics:

$$\mathbf{B}[\mathbf{t}] - B_{\text{prev}}[t] + \Delta t \cdot P_B[t] = 0 \quad (13)$$

where:

$$B_{\text{prev}}[t] = \begin{cases} B_0 & \text{if } t = 1 \\ \mathbf{B}^k[\mathbf{t} - \mathbf{1}] & \text{if } t > 1 \quad (\text{from current forward pass}) \end{cases} \quad (14)$$

SOC Box Constraints (Negative Inequality Form):

$$\underline{B} - \mathbf{B}[\mathbf{t}] \leq 0 \quad (\text{i.e., } \mathbf{B}[\mathbf{t}] \geq \underline{B}) \quad (15)$$

$$\mathbf{B}[\mathbf{t}] - \overline{B} \leq 0 \quad (\text{i.e., } \mathbf{B}[\mathbf{t}] \leq \overline{B}) \quad (16)$$

Battery Power Limits:

$$-P_{B,R} \leq P_B[t] \leq P_{B,R} \quad (17)$$

$$-P_{B,R} \leq P_B[t+1] \leq P_{B,R} \quad (\text{if } t < T) \quad (18)$$

Terminal SOC Constraint (if $t = T$ and terminal constraint exists):

$$\mathbf{B}[\mathbf{T}] = B_{T,\text{target}} \quad (19)$$

4.2 Key Formulation Insights

4.2.1 Coupling Structure

- **Backward (state) coupling:** Each subproblem uses $\mathbf{B}[\mathbf{t} - \mathbf{1}]$ from the *current* forward pass (iteration k), ensuring consistency within the same pass.
- **Forward (cost-to-go) coupling:** The coupling term with time $t + 1$ uses:
 - $\mathbf{B}^{k-1}[\mathbf{t} + \mathbf{1}]$: State from *previous* iteration (parameter)
 - $\mu^{k-1}[\mathbf{t} + \mathbf{1}]$: Shadow price from *previous* iteration (parameter)
 - $P_B[t + 1]$: Battery power at $t + 1$ as a *decision variable*
- The coupling term $\mu[\mathbf{t} + \mathbf{1}] \cdot (\mathbf{B}[\mathbf{t} + \mathbf{1}] - \mathbf{B}[\mathbf{t}] + \Delta t \cdot P_B[t + 1])$ penalizes violations of the battery dynamics constraint between t and $t + 1$.
- By optimizing $P_B[t + 1]$ as a decision variable, each subproblem can "soft-enforce" consistency with the next time step, even though $\mathbf{B}[\mathbf{t} + \mathbf{1}]$ is fixed from the previous iteration.

4.2.2 Shadow Price Interpretation

The dual variable $\mu[\mathbf{t}]$ from the battery dynamics constraint represents:

- Marginal value of increasing $\mathbf{B}[\mathbf{t}]$ by one unit
- Shadow price / cost-to-go from state $\mathbf{B}[\mathbf{t}]$ to terminal time T
- How much the objective would improve if we had one more unit of energy stored at time t

For terminal time step $t = T$ with terminal constraint $\mathbf{B}[\mathbf{T}] = B_{T,\text{target}}$:

$$\mu_{\text{total}}[\mathbf{T}] = \mu_{\text{dynamics}}[\mathbf{T}] + \mu_{\text{terminal}}[\mathbf{T}] \quad (20)$$

where $\mu_{\text{terminal}}[\mathbf{T}]$ is the dual of the terminal constraint.

4.2.3 No Explicit Dual Update

Unlike ADMM-based methods, DDP does *not* require a separate dual update step. The shadow prices $\mu[t]$ are automatically extracted from each subproblem's battery dynamics constraint dual:

$$\mu[t] = -\lambda_{\text{battery_dynamics}}[t] \quad (21)$$

where $\lambda_{\text{battery_dynamics}}[t]$ is the JuMP dual variable (negated for proper economic sign convention).

5 DDP Algorithm

Algorithm 1 DDP Forward Pass for Copper Plate MPOPF

```

1: Input: Problem data  $\mathcal{D} = \{T, C[t], P_L[t], B_0, E_{\text{Rated}}, P_{B,R}, \text{SOC}_{\min}, \text{SOC}_{\max}, \Delta t\}$ 
2: Parameters: max_iter,  $\epsilon_{\text{tol}}$ , optional  $\mu^0$  (initial dual variables)
3:
4: Initialize ( $k = 0$ ):
5:    $\mathbf{B}^0[t] \leftarrow B_0$  for all  $t \in \mathcal{T}$ 
6:    $\mu^0[t] \leftarrow 0$  (or provided warm start) for all  $t \in \mathcal{T}$ 
7:    $P_B^0[t] \leftarrow 0, P_{\text{subs}}^0[t] \leftarrow 0$  for all  $t \in \mathcal{T}$ 
8:
9: for  $k = 1, 2, \dots, \text{max\_iter}$  do
10:  // Store previous iteration values for coupling
11:   $\mathbf{B}^{k-1} \leftarrow \mathbf{B}^{k-1}, \mu^{k-1} \leftarrow \mu^{k-1}, P_B^{k-1} \leftarrow P_B^{k-1}$ 
12:
13:  // Forward Pass: Solve each time step sequentially
14:  for  $t = 1, 2, \dots, T$  do
15:    Solve subproblem for time  $t$  using:
16:     $\mathbf{B}^k[t-1]$  from current forward pass (if  $t > 1$ )
17:     $\mathbf{B}^{k-1}[t+1], \mu^{k-1}[t+1]$  from previous iteration (if  $t < T$ )
18:
19:    Extract solution:
20:     $\mathbf{B}^k[t] \leftarrow$  optimal SOC
21:     $P_B^k[t] \leftarrow$  optimal battery power
22:     $P_{\text{subs}}^k[t] \leftarrow$  optimal substation power
23:     $\mu^k[t] \leftarrow -\text{dual}(\text{battery\_dynamics}[t])$ 
24:    If  $t = T$  with terminal constraint:
25:       $\mu^k[T] \leftarrow \mu^k[T] + (-\text{dual}(\text{terminal\_soc}))$ 
26:
27:    if  $t < T$  then
28:       $P_B^k[t+1] \leftarrow$  optimal value of  $P_B[t+1]$  variable
29:    end if
30:  end for
31:
32:  // Compute convergence metrics
33:   $\text{err}_{\text{state}} \leftarrow \|\mathbf{B}^k - \mathbf{B}^{k-1}\|_2$ 
34:   $\text{err}_{\text{dual}} \leftarrow \|\mu^k - \mu^{k-1}\|_2$ 
35:   $\text{err}_{\text{KKT}} \leftarrow \max(\text{err}_{\text{state}}, \text{err}_{\text{dual}})$ 
36:
37:  if  $\text{err}_{\text{KKT}} < \epsilon_{\text{tol}}$  then
38:    break ▷ Converged
39:  end if
40: end for
41:
42: Return:  $\{\mathbf{B}^k[t], P_B^k[t], P_{\text{subs}}^k[t], \mu^k[t]\}_{t=1}^T$ , convergence status

```

6 Convergence Criteria

6.1 State Trajectory Error

$$\text{err}_{\text{state}}^k = \left\| \mathbf{B}^k - \mathbf{B}^{k-1} \right\|_2 = \sqrt{\sum_{t=1}^T (\mathbf{B}^k[t] - \mathbf{B}^{k-1}[t])^2} \quad (22)$$

6.2 Dual Variable Error

$$\text{err}_{\text{dual}}^k = \left\| \mu^k - \mu^{k-1} \right\|_2 = \sqrt{\sum_{t=1}^T (\mu^k[t] - \mu^{k-1}[t])^2} \quad (23)$$

6.3 KKT Error (Combined)

$$\text{err}_{\text{KKT}}^k = \max \left(\text{err}_{\text{state}}^k, \text{err}_{\text{dual}}^k \right) \quad (24)$$

The algorithm converges when:

$$\text{err}_{\text{KKT}}^k < \epsilon_{\text{tol}} \quad (\text{typically } \epsilon_{\text{tol}} = 10^{-5}) \quad (25)$$

7 Objective Function

7.1 Total Objective (Physical Units)

$$J = \sum_{t=1}^T [C[t] \cdot (P_{\text{subs}}[t] \cdot P_{\text{BASE}}) \cdot \Delta t + C_B \cdot (P_B[t] \cdot P_{\text{BASE}})^2 \cdot \Delta t] \quad (26)$$

7.2 Objective Components

$$J_{\text{energy}} = \sum_{t=1}^T C[t] \cdot (P_{\text{subs}}[t] \cdot P_{\text{BASE}}) \cdot \Delta t \quad [\$] \quad (27)$$

$$J_{\text{battery}} = \sum_{t=1}^T C_B \cdot (P_B[t] \cdot P_{\text{BASE}})^2 \cdot \Delta t \quad [\$] \quad (28)$$

$$J = J_{\text{energy}} + J_{\text{battery}} \quad [\$] \quad (29)$$

7.3 Battery Quadratic Cost Rationale

The battery cost term $C_B \cdot (P_B[t])^2$ serves multiple purposes:

1. **Regularization:** Prevents numerical issues and excessive battery cycling
2. **Smoothing:** Encourages gradual power transitions rather than bang-bang control
3. **Conditioning:** Improves numerical stability of the optimization problem
4. **Physical realism:** Approximates wear-and-tear and efficiency losses

Typically set as: $C_B = 10^{-6} \times \min(C[t])$ to ensure it's a small regularization term.

8 Dual Variable Extraction and Sign Conventions

8.1 Battery Dynamics Dual

From the constraint:

$$\mathbf{B}[t] - B_{\text{prev}}[t] + \Delta t \cdot P_B[t] = 0 \quad (30)$$

JuMP returns the dual $\lambda_{\text{dynamics}}[t]$. The shadow price is:

$$\mu_{\text{dynamics}}[t] = -\lambda_{\text{dynamics}}[t] \quad (31)$$

Economic interpretation: $\mu[t] > 0$ means increasing $\mathbf{B}[t]$ would *decrease* the objective (energy stored is valuable). $\mu[t] < 0$ means increasing $\mathbf{B}[t]$ would *increase* the objective (energy stored is costly).

8.2 SOC Box Constraint Duals

From constraints (negative inequality form):

$$g_{\text{Bmin}}[t] = \underline{B} - \mathbf{B}[t] \leq 0 \quad (\text{i.e., } \mathbf{B}[t] \geq \underline{B}) \quad (32)$$

$$g_{\text{Bmax}}[t] = \mathbf{B}[t] - \overline{B} \leq 0 \quad (\text{i.e., } \mathbf{B}[t] \leq \overline{B}) \quad (33)$$

JuMP dual variables:

$$\lambda_{\text{Bmin}}[t] = -\text{dual}(g_{\text{Bmin}}[t]) \quad (\text{negate to get positive when active}) \quad (34)$$

$$\lambda_{\text{Bmax}}[t] = \text{dual}(g_{\text{Bmax}}[t]) \quad (\text{already correct sign}) \quad (35)$$

Complementary slackness:

$$\lambda_{\text{Bmin}}[t] \cdot (\mathbf{B}[t] - \underline{B}) = 0, \quad \lambda_{\text{Bmin}}[t] \geq 0 \quad (36)$$

$$\lambda_{\text{Bmax}}[t] \cdot (\overline{B} - \mathbf{B}[t]) = 0, \quad \lambda_{\text{Bmax}}[t] \geq 0 \quad (37)$$

8.3 Terminal SOC Constraint Dual

If terminal constraint exists: $\mathbf{B}[\mathbf{T}] = B_{T,\text{target}}$

The total shadow price at $t = T$ is:

$$\mu_{\text{total}}[\mathbf{T}] = \mu_{\text{dynamics}}[\mathbf{T}] + \mu_{\text{terminal}}[\mathbf{T}] \quad (38)$$

where:

$$\mu_{\text{terminal}}[\mathbf{T}] = -\text{dual}(\text{terminal_soc}) \quad (39)$$

9 KKT Optimality Conditions

For each time step t , the KKT conditions are:

9.1 Stationarity

$$\frac{\partial \mathcal{L}}{\partial P_{\text{subs}}[t]} = C[t] \cdot \Delta t + \lambda_{\text{power_balance}}[t] = 0 \quad (40)$$

$$\frac{\partial \mathcal{L}}{\partial P_B[t]} = 2C_B \cdot P_B[t] \cdot \Delta t + \lambda_{\text{power_balance}}[t] + \Delta t \cdot \mu[\mathbf{t}] = 0 \quad (41)$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{B}[\mathbf{t}]} = \mu[\mathbf{t}] - \mu[\mathbf{t} + \mathbf{1}] + \lambda_{\text{Bmin}}[t] - \lambda_{\text{Bmax}}[t] = 0 \quad (42)$$

For terminal step $t = T$ with terminal constraint:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{B}[\mathbf{T}]} = \mu_{\text{dynamics}}[\mathbf{T}] + \mu_{\text{terminal}}[\mathbf{T}] + \lambda_{\text{Bmin}}[T] - \lambda_{\text{Bmax}}[T] = 0 \quad (43)$$

9.2 Primal Feasibility

All constraints satisfied:

$$P_{\text{subs}}[t] + P_B[t] - P_L[t] = 0 \quad (44)$$

$$\mathbf{B}[\mathbf{t}] - B_{\text{prev}}[t] + \Delta t \cdot P_B[t] = 0 \quad (45)$$

$$\underline{B} \leq \mathbf{B}[\mathbf{t}] \leq \overline{B} \quad (46)$$

$$-P_{B,R} \leq P_B[t] \leq P_{B,R} \quad (47)$$

9.3 Dual Feasibility

$$\lambda_{\text{Bmin}}[t] \geq 0 \quad (48)$$

$$\lambda_{\text{Bmax}}[t] \geq 0 \quad (49)$$

9.4 Complementary Slackness

$$\lambda_{\text{Bmin}}[t] \cdot (\mathbf{B}[\mathbf{t}] - \underline{B}) = 0 \quad (50)$$

$$\lambda_{\text{Bmax}}[t] \cdot (\overline{B} - \mathbf{B}[\mathbf{t}]) = 0 \quad (51)$$

10 Comparison with Brute Force Solution

10.1 Centralized (Brute Force) Formulation

The brute force approach solves the full multi-period problem in one shot:

$$\min_{\{P_{\text{subs}}[t], P_B[t], B[t]\}_{t=1}^T} \sum_{t=1}^T [C[t] \cdot P_{\text{subs}}[t] \cdot \Delta t + C_B \cdot (P_B[t])^2 \cdot \Delta t] \quad (52)$$

Subject to:

$$P_{\text{subs}}[t] + P_B[t] = P_L[t], \quad \forall t \in \mathcal{T} \quad (53)$$

$$B[1] - B_0 + \Delta t \cdot P_B[1] = 0 \quad (54)$$

$$B[t] - B[t-1] + \Delta t \cdot P_B[t] = 0, \quad \forall t \in \{2, \dots, T\} \quad (55)$$

$$\underline{B} \leq B[t] \leq \overline{B}, \quad \forall t \in \mathcal{T} \quad (56)$$

$$-P_{B,R} \leq P_B[t] \leq P_{B,R}, \quad \forall t \in \mathcal{T} \quad (57)$$

$$B[T] = B_{T,\text{target}} \quad (\text{if terminal constraint exists}) \quad (58)$$

10.2 Key Differences

- **Brute Force:** Solves all time steps simultaneously, guarantees global optimum
- **DDP:** Solves time steps sequentially with iterative coupling, may require multiple passes to converge
- **Scalability:** DDP can parallelize across time steps (not implemented in current version)
- **Memory:** DDP has smaller subproblems, lower memory footprint per solve
- **Warm starting:** DDP can initialize with duals from brute force for rapid convergence

11 Convergence Properties

11.1 Optimality Guarantee

Under convexity assumptions (which hold for the copper plate problem):

- If DDP converges (i.e., $\text{err}_{\text{KKT}} < \epsilon_{\text{tol}}$), the solution is **globally optimal**
- Convergence is guaranteed if the coupling penalty (implicitly through dual variables) is strong enough
- With warm start from brute force duals, DDP often converges in 1-2 iterations

11.2 Convergence Rate

- **Linear convergence:** $\text{err}^{k+1} \leq \alpha \cdot \text{err}^k$ for some $\alpha < 1$
- Convergence rate depends on problem conditioning and coupling strength
- Tighter SOC bounds $\Rightarrow$ slower convergence (more binding constraints)
- More time periods $T \Rightarrow$ longer horizon coupling, potentially slower convergence

11.3 Practical Observations

From numerical experiments with $T = 2, 4$ time periods:

- **Cold start** ($\mu = 0$): Converges in 5-15 iterations to $\epsilon_{\text{tol}} = 10^{-5}$
- **Warm start** (μ from BF): Converges in 1-2 iterations
- **Objective accuracy:** Matches brute force to within 10^{-6} relative error
- **Solution accuracy:** State and control trajectories match BF to numerical precision

12 Implementation Notes

12.1 Solver Choices

- **Gurobi**: Fast for quadratic programs (QP), commercial license required
- **Ipopt**: Open-source nonlinear solver, slightly slower but freely available
- Both solvers produce identical solutions for this convex problem

12.2 Numerical Considerations

- Use per-unit system to avoid ill-conditioning from large physical values
- Set C_B small enough to be a regularization, not a dominant cost
- Terminal SOC constraint should not be at SOC bounds to avoid numerical issues
- Monitor dual variable magnitudes to detect potential infeasibility

12.3 Parallelization Potential

Current implementation is *sequential* forward pass. Future extensions:

- Parallelize subproblems within each iteration (requires asynchronous updates)
- Use GPU acceleration for large-scale systems
- Implement backward pass for improved convergence

13 Example Problem Instance

13.1 System Configuration

$$T = 4 \text{ time periods} \quad (59)$$

$$\Delta t = 1.0 \text{ hour} \quad (60)$$

$$P_{\text{BASE}} = 1000 \text{ kW} \quad (61)$$

$$E_{\text{BASE}} = 1000 \text{ kWh} \quad (62)$$

13.2 Battery Specifications

$$E_{\text{Rated}} = 4000 \text{ kWh} = 4.0 \text{ pu} \quad (63)$$

$$P_{B,R} = 800 \text{ kW} = 0.8 \text{ pu} \quad (64)$$

$$\text{SOC}_{\min} = 0.30 \quad (30\%) \quad (65)$$

$$\text{SOC}_{\max} = 0.95 \quad (95\%) \quad (66)$$

$$B_0 = 0.30 \times 4.0 = 1.2 \text{ pu} \quad (30\% \text{ initial SOC}) \quad (67)$$

$$B_{T,\text{target}} = 0.32 \times 4.0 = 1.28 \text{ pu} \quad (32\% \text{ terminal SOC}) \quad (68)$$

13.3 Load Profile (Rated = 1250 kW)

$$P_L[t] = 1250 \times \text{LoadShape}[t]/1000 \quad [\text{pu}] \quad (69)$$

$$\text{LoadShape} = 0.8 + 0.2 \times \frac{\sin(\omega t - 0.8) + 1}{2}, \quad \omega = \frac{2\pi}{T} \quad (70)$$

Example for $T = 4$: $P_L \approx [1.00, 1.10, 1.15, 1.05]$ pu (approximate sinusoidal)

13.4 Price Profile (Charge-Discharge Scenario)

Design to test look-ahead capability:

$$C[t] = \begin{cases} 0.05 \text{ \$/kWh} & t \in \{1, 2\} \quad (\text{cheap: charge}) \\ 0.20 \text{ \$/kWh} & t \in \{3, 4\} \quad (\text{expensive: discharge}) \end{cases} \quad (71)$$

This creates a $4\times$ price difference to incentivize:

- **Periods 1-2:** Charge battery (low prices)
- **Periods 3-4:** Discharge battery (high prices)

13.5 Battery Cost Coefficient

$$C_B = 10^{-6} \times \min(C[t]) = 10^{-6} \times 0.05 = 5 \times 10^{-8} \text{ \$/kW}^2/\text{h} \quad (72)$$

14 Appendix: Physical Interpretation

14.1 Battery Power Sign Convention

- $P_B[t] > 0$: Battery **discharging** (injecting power into grid)
- $P_B[t] < 0$: Battery **charging** (consuming power from grid)
- $P_B[t] = 0$: Battery idle

14.2 Economic Signals

- High $C[t] \Rightarrow$ Expensive energy $\Rightarrow$ Discharge battery ($P_B[t] > 0$) $\Rightarrow$ Reduce $P_{\text{subs}}[t]$
- Low $C[t] \Rightarrow$ Cheap energy $\Rightarrow$ Charge battery ($P_B[t] < 0$) $\Rightarrow$ Increase $P_{\text{subs}}[t]$
- $\mu[t] > 0 \Rightarrow$ Energy stored at t is valuable (future high prices)
- $\mu[t] < 0 \Rightarrow$ Energy stored at t is costly (future low prices)

14.3 SOC Trajectory Interpretation

- $B[t]$ decreases when $P_B[t] > 0$ (discharging)
- $B[t]$ increases when $P_B[t] < 0$ (charging)
- Optimal trajectory balances energy arbitrage profit against SOC constraints
- Terminal constraint $B[T] = B_{T,\text{target}}$ ensures battery returns to desired state